# Mission Report - Team Rocket

## Introduction

This report describes the digital signal processing (DSP) pipeline used to recover telemetry data from the Voyager-X probe's backup beacon signal. The Deep Space Network captured a weak radio signal embedded in noise. Our objective was to detect the signal, estimate the carrier frequency, correct the carrier offset, recover symbol timing, identify the modulation scheme, and extract the transmitted data.

## 1. Handling Memory Constraints (~15 GB Dataset)

The dataset consisted of approximately 300 text files containing hexadecimal IEEE-754 floating-point IQ samples, totaling nearly 15 GB of raw signal data. Loading the entire dataset simultaneously would require tens of gigabytes of RAM after conversion to complex numbers.

To address this, a stream-based processing approach was used. Files were processed sequentially and data was read line-by-line, converting hexadecimal values directly into complex samples

$s[n] = I[n] + jQ[n]$

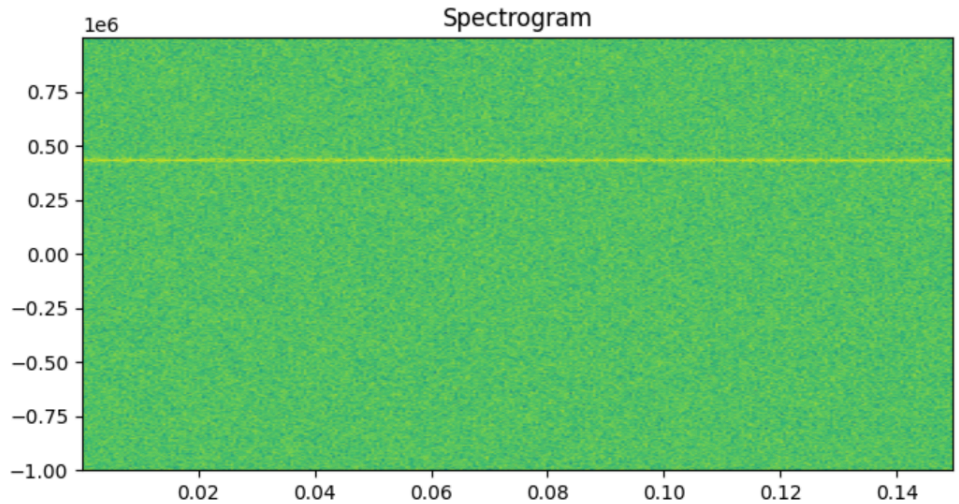
where $I[n]$ and $Q[n]$ represent the in-phase and quadrature components.

This approach ensured that only a small portion of the dataset remained in memory at any time, enabling efficient analysis of large signal recordings.

## 2. Spectrogram / Waterfall Analysis

A spectrogram (short-time Fourier transform) was used to detect the presence of the signal and observe frequency behavior over time.

A faint spectral line revealed the carrier around ~430 kHz above the noise floor.




The received signal is represented as a complex baseband signal:

$s[n] = I[n] + jQ[n]$

Relationship between sample index and time:

$t_n = n / F_s$

Spectral analysis uses the Discrete Fourier Transform (DFT):

$S[k] = \Sigma\, s[n]\, e^{(-j2\pi kn/N)}$

Power spectrum:

$P[k] = |S[k]|^2$

## 3. Carrier Detection and Frequency Translation

The carrier frequency was estimated by locating the frequency with maximum spectral magnitude

$f_c = \text{argmax}\, |S(f)|$

After detecting the carrier, the signal was translated to baseband using:

$s_{bb}(n) = s(n)\, e^{(-j2\pi f_c\, n / F_s)}$

This operation removes the carrier offset and centers the signal spectrum around 0 Hz, simplifying subsequent demodulation.

## 4. Digital Filtering

A digital Butterworth low-pass filter removes noise outside signal bandwidth.

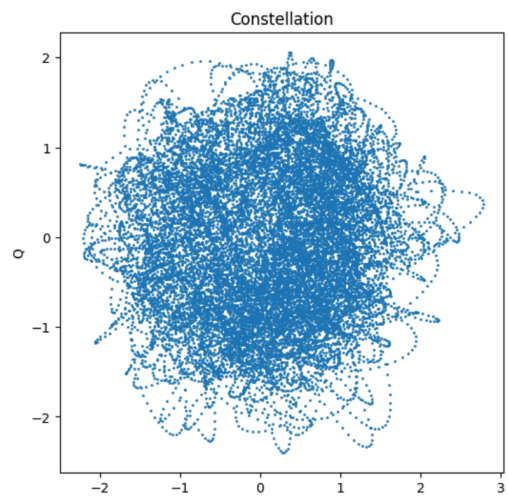
General filter equation:

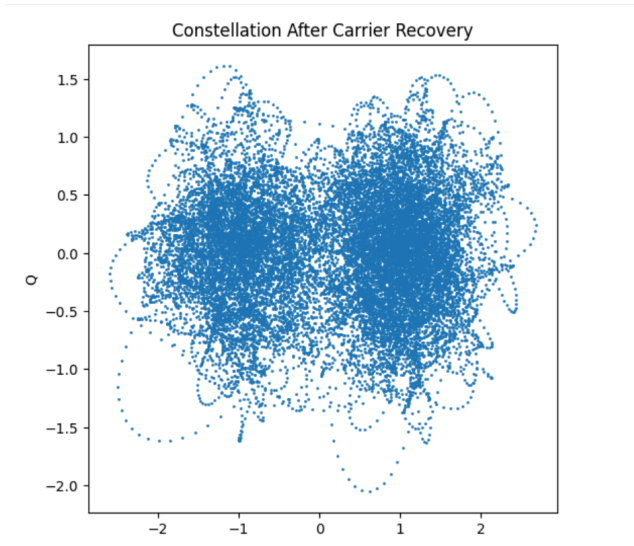
$$y[n] = \Sigma\, b_k\, x[n-k] - \Sigma\, a_k\, y[n-k]$$

## 5. Carrier Recovery and Constellation Analysis

After detecting the carrier, the signal was shifted to baseband by multiplying the signal with a complex exponential. This removed the carrier frequency offset and allowed us to analyze the modulation constellation in the I-Q plane.

The initial constellation appeared as a noisy circular cloud due to phase drift. After carrier correction, two clear clusters became visible, indicating that the modulation scheme is Binary Phase Shift Keying (BPSK).






Each received symbol can be represented as:

$s = A\, e^{(j\varphi)}$

For Binary Phase Shift Keying (BPSK):

$s \in \{+A, -A\}$

## 6. Symbol Timing Recovery

Autocorrelation and envelope spectrum analysis were used to estimate the symbol rate. The envelope spectrum revealed a clear peak corresponding to approximately 50,000 symbols per second. This allowed proper sampling of the received signal to recover symbol decisions.

```
sr = symbol_rate

sps = FS / sr
```

Samples per symbol: 49

Symbol timing is estimated using autocorrelation:

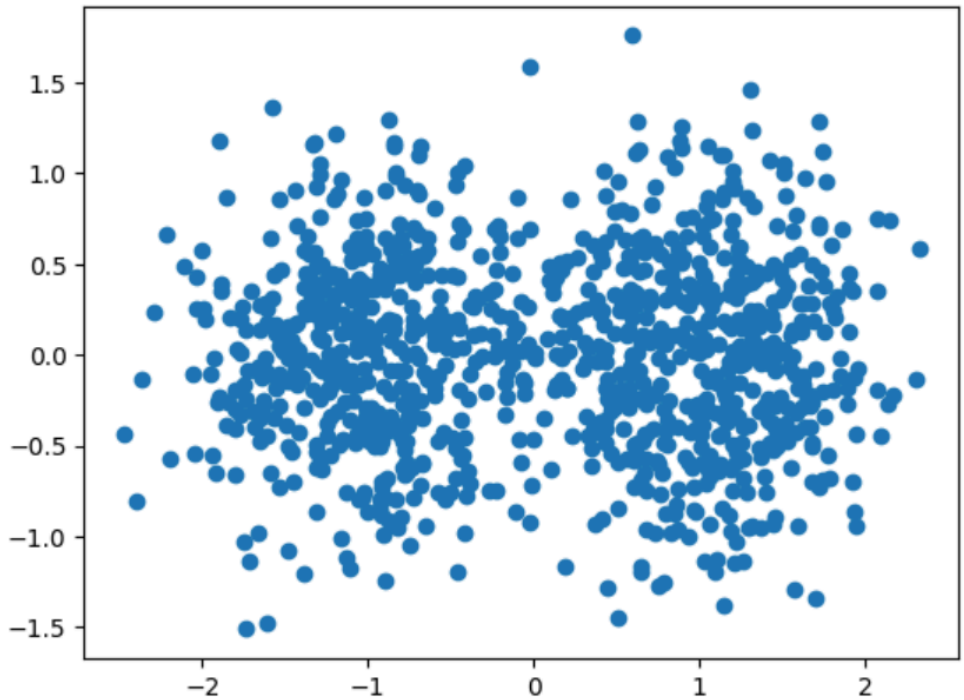
$R(\tau) = \Sigma\, s[n]\, s^*(n - \tau)$

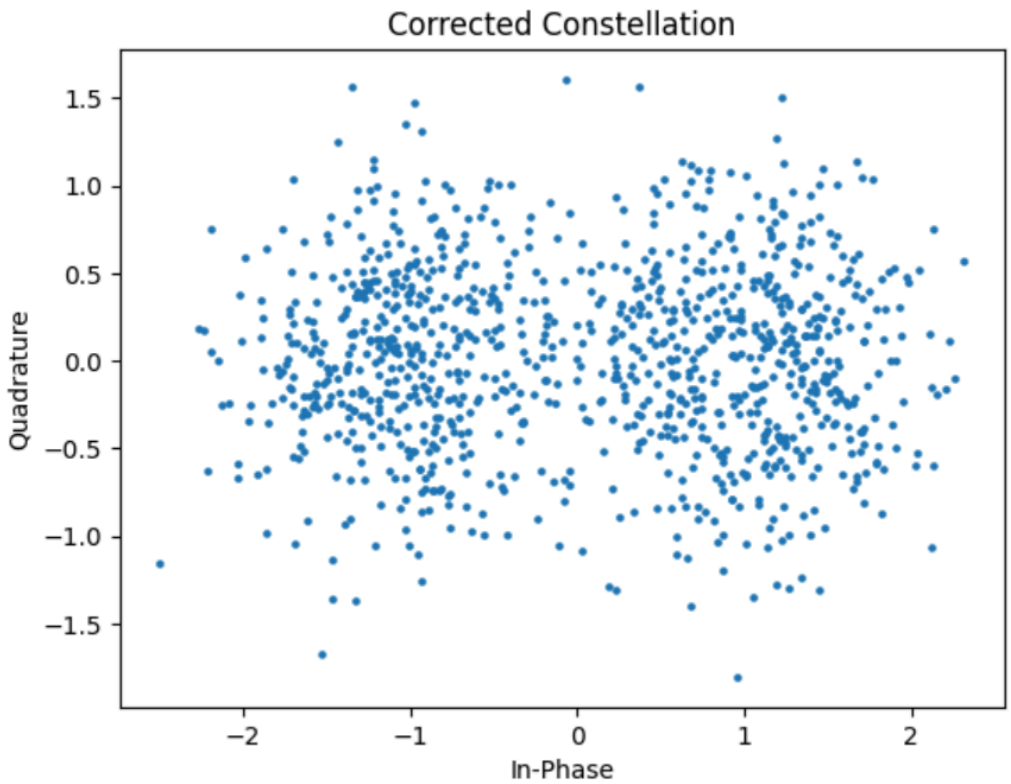
Envelope signal:

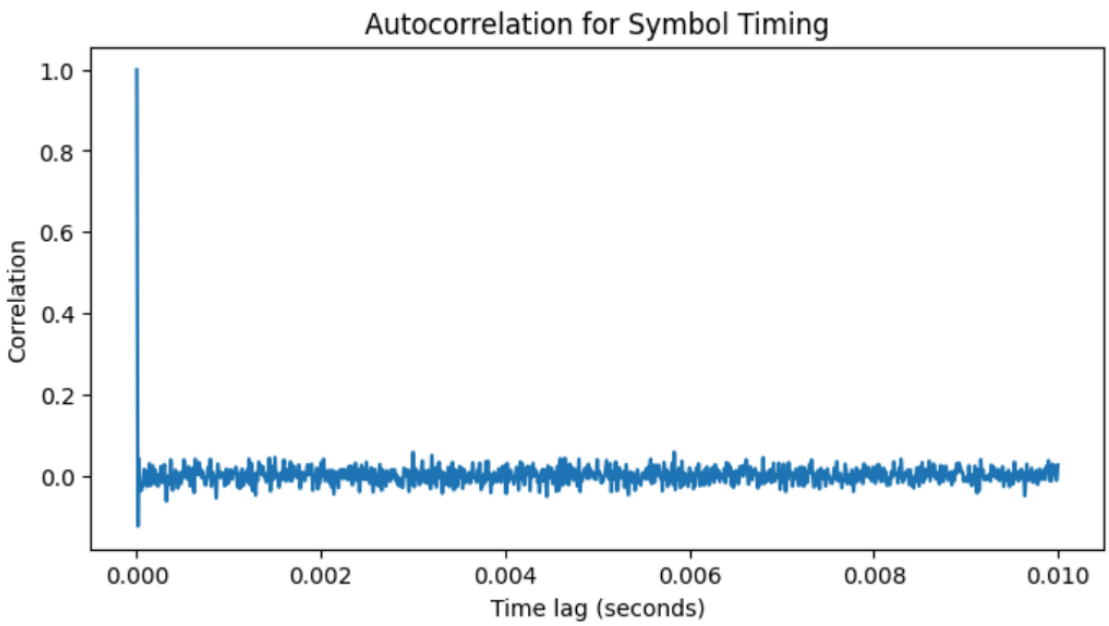
E[n] = |s[n]|

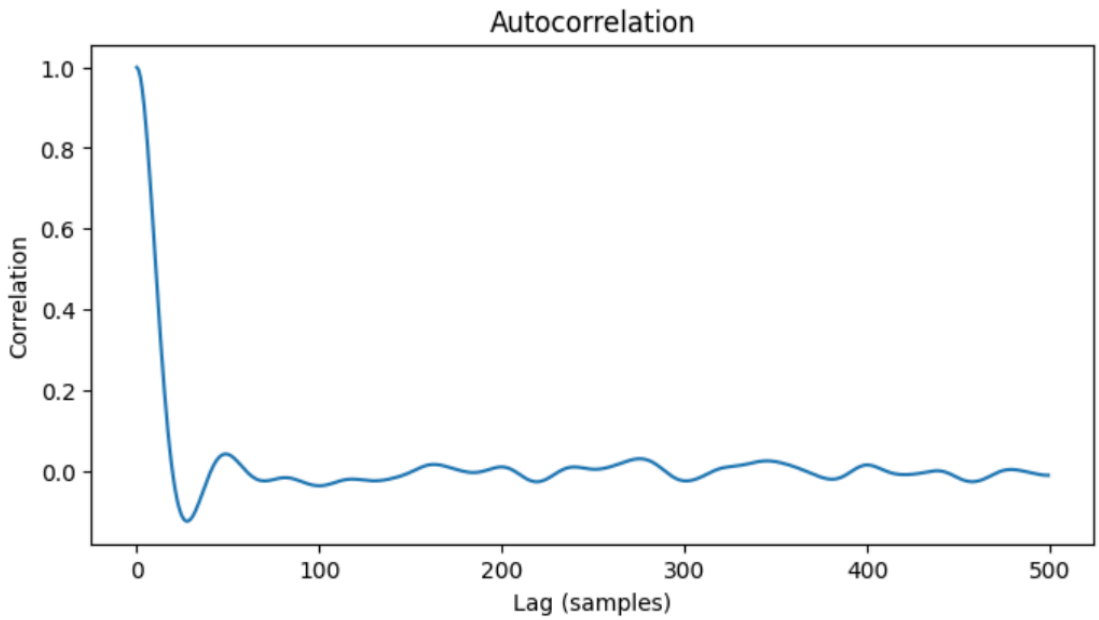
Symbol rate estimate:

R_s ≈ 50,000 symbols/s




Corrected Constellation
Quadrature
In-Phase
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2
−1
0
1
2


Autocorrelation for Symbol Timing
Correlation
Time lag (seconds)
1.0
0.8
0.6
0.4
0.2
0.0
0.000
0.002
0.004
0.006
0.008
0.010




## 7. BPSK Demodulation

Binary Phase Shift Keying (BPSK) is a digital modulation technique where binary data is represented using two phase states of the carrier signal, typically 0° and 180°. During demodulation, after carrier recovery, the received symbols are examined on the real axis: symbols with positive real values are interpreted as bit 1, while negative real values are interpreted as bit 0. This decision rule allows recovery of the transmitted binary data.

Bit decision rule for BPSK:

bit = 1 if I > 0, bit = 0 if I < 0

I - in phase symbol

## 8. CCSDS Synchronization Word:

The Consultative Committee for Space Data Systems (CCSDS) defines standard packet formats for satellite communication. Each telemetry frame begins with a fixed 32-bit synchronization word (0x1ACFFC1D), which helps the receiver identify the start of a valid packet in the received bitstream. By correlating the received bits with this known pattern, the decoder can locate packet boundaries and extract the header and payload data correctly.

The packets found using this method are:

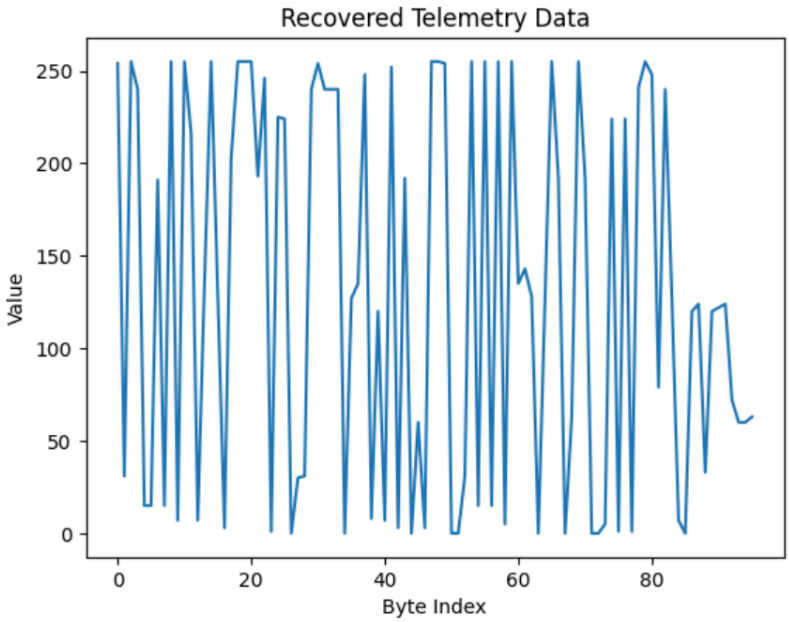
Packet 1

Bit position: 197

Header: [3, 255, 252, 31, 254, 30]

Payload [3, 255, 252, 31, 254, 30, 254, 31, 255, 240, 15, 15, 191, 15, 255, 7, 255, 216, 7, 135, 255, 128, 3, 203, 255, 255, 255, 193, 246, 1, 225, 224, 0, 30, 31, 240, 254, 240, 240, 240, 0, 127, 135, 248, 8, 120, 7, 252, 3, 192, 0, 60, 3, 255, 255, 254, 0, 0, 31, 255, 15, 255, 15, 255, 5, 255, 135, 143, 128, 0, 127, 255, 192, 0, 63, 255, 192, 0, 0, 5, 224, 1, 224, 1, 241, 255, 248, 79, 240, 127, 7, 0, 120, 124, 33, 120, 122, 124, 72, 60, 60, 63]

Telemetry Payload : [254, 31, 255, 240, 15, 15, 191, 15, 255, 7, 255, 216, 7, 135, 255, 128, 3, 203, 255, 255, 255, 193, 246, 1, 225, 224, 0, 30, 31, 240, 254, 240, 240, 240, 0, 127, 135, 248, 8, 120, 7, 252, 3, 192, 0, 60, 3, 255, 255, 254, 0, 0, 31, 255, 15, 255, 15, 255, 5, 255, 135, 143, 128, 0, 127, 255, 192, 0, 63, 255, 192, 0, 0, 5, 224, 1, 224, 1, 241, 255, 248, 79, 240, 127, 7, 0, 120, 124, 33, 120, 122, 124, 72, 60, 60, 63]



Telemetry Value Reconstruction

The telemetry payload consists of 8-bit bytes. To obtain meaningful telemetry measurements, two consecutive bytes are combined to form a 16-bit value. This is done by shifting the first byte to the higher 8 bits and combining it with the second byte using a bitwise OR operation.

$$T = (B1 << 8) \mid B2$$

where:

B1= first byte

B2= second byte

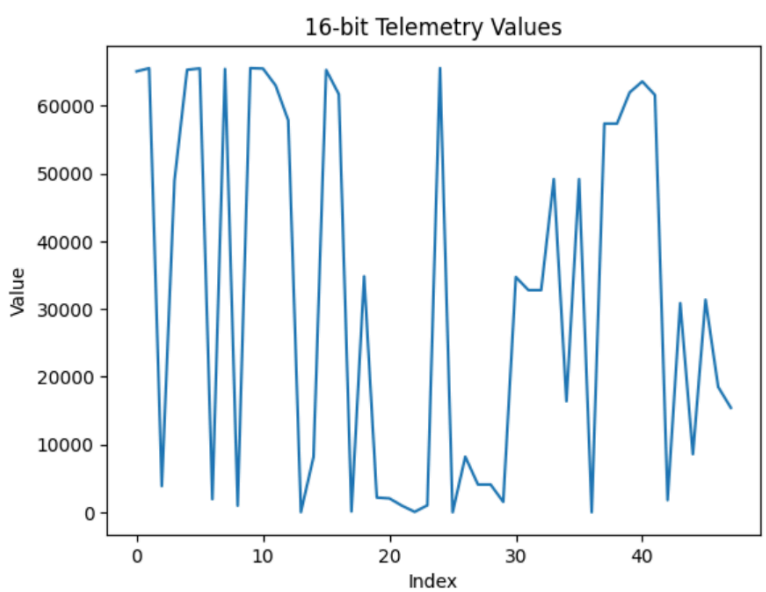
<<= left bit shift

| = bitwise OR operation

T = reconstructed 16-bit telemetry value

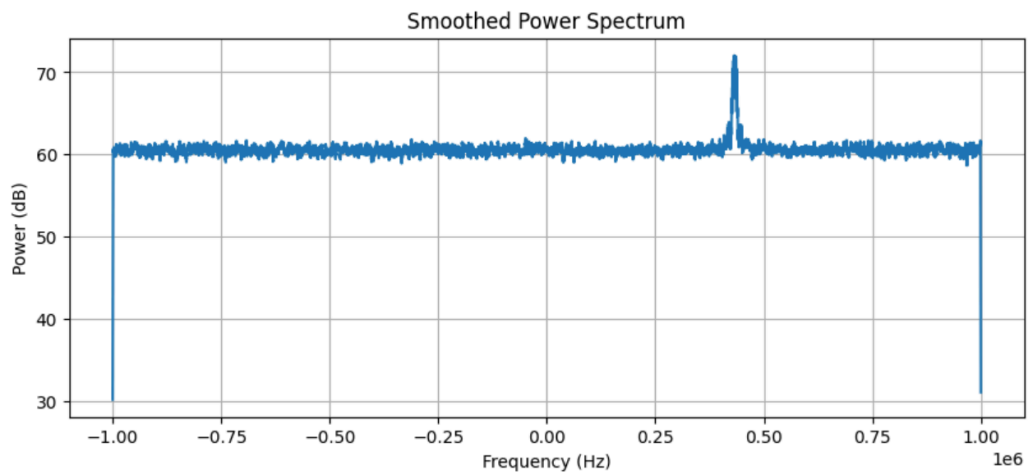
This process converts two 8-bit bytes into a single 16-bit telemetry parameter, which can represent spacecraft sensor readings or system status data.

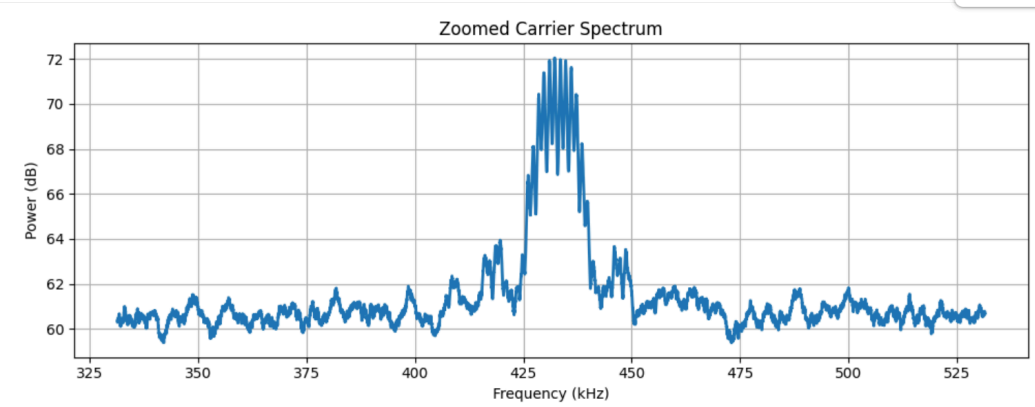
telemetry values are : [65055, 65520, 3855, 48911, 65287, 65496, 1927, 65408, 971, 65535, 65473, 62977, 57824, 30, 8176, 65264, 61680, 127, 34808, 2168, 2044, 960, 60, 1023, 65534, 0, 8191, 4095, 4095, 1535, 34703, 32768, 32767, 49152, 16383, 49152, 5, 57345, 57345, 61951, 63567, 61567, 1792, 30844, 8568, 31356, 18492, 15423]

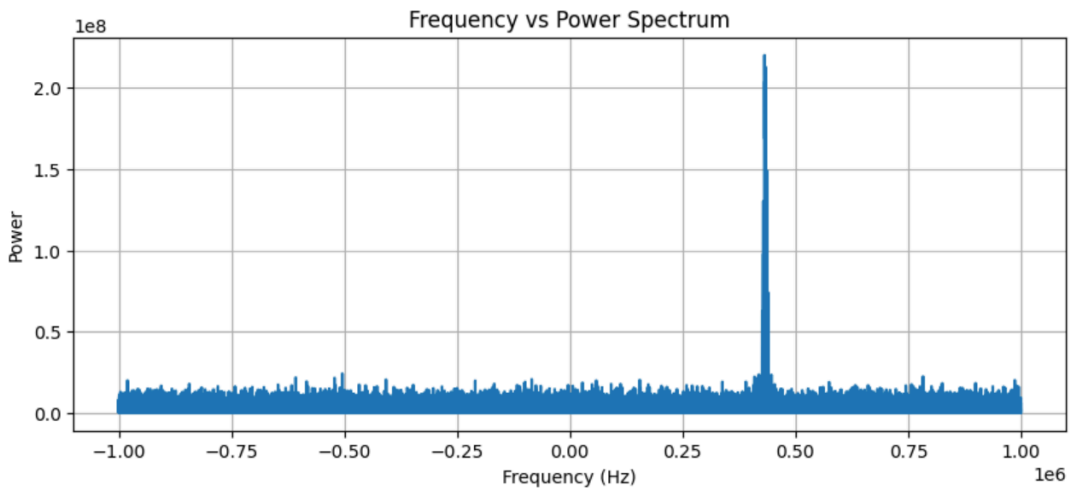


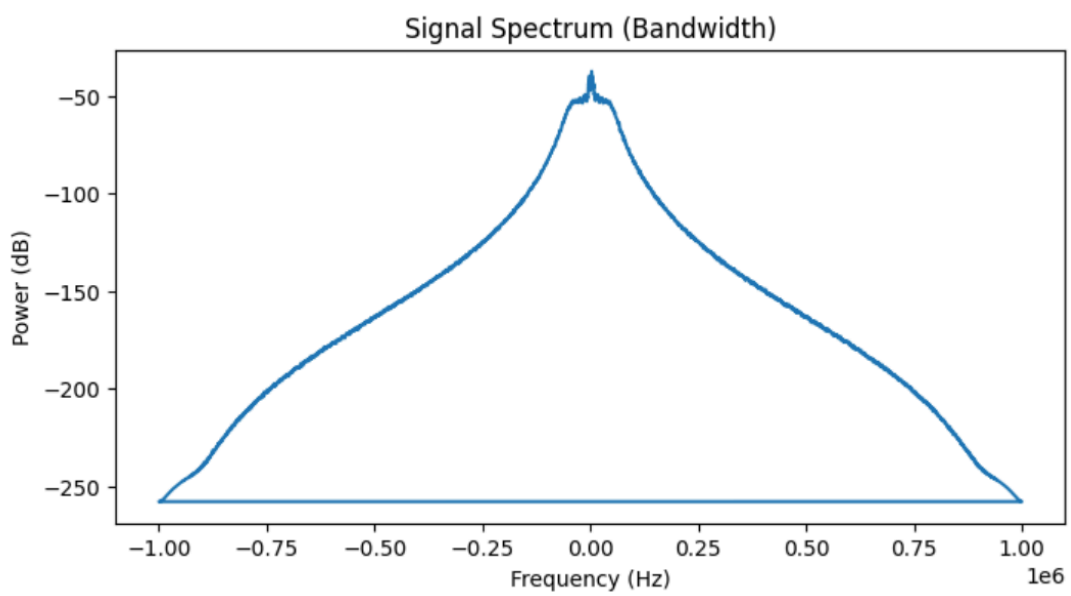
## 9. Data-Layer Processing

After symbol decisions were obtained, the resulting bitstream was analyzed for packet synchronization patterns. The CCSDS Attached Sync Marker (ASM) 0x1ACFFC1D was searched within the bitstream to identify frame boundaries. Once frames were aligned, payload data could be extracted for further analysis.






## Signal Processing Pipeline

Raw IQ Dataset

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Spectral Analysis

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Carrier Detection

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Baseband Conversion

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Filtering

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Symbol Timing Recovery

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BPSK Demodulation

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Telemetry Packet Detection

## Conclusion

The implemented DSP pipeline successfully detected and recovered the Voyager-X beacon signal from a noisy dataset. Carrier detection, baseband conversion, symbol timing estimation, and constellation analysis confirmed a BPSK-modulated telemetry signal with an estimated symbol rate of approximately 50 kSymbols/s.